

Kingdom of Science: physics.pdf

Levers & Torque

Torque equilibrium ($\text{Torque} = \text{Force} \times \text{Distance}$). Required: Timber beam, fulcrum stone. Procedure: Place fulcrum close to heavy load, apply downward effort at long end of beam. Safety: Ensure fulcrum stability.

Archimedes Screw Water Pump

Helical blade inside cylinder lifts water when rotated. Required: Wooden axle pole, bamboo/metal helix, pipe casing, crank. Procedure: Wrap helix around pole at 45 deg, enclose in casing, submerge intake end at 30 deg incline and rotate handle. Safety: Secure bank anchor.